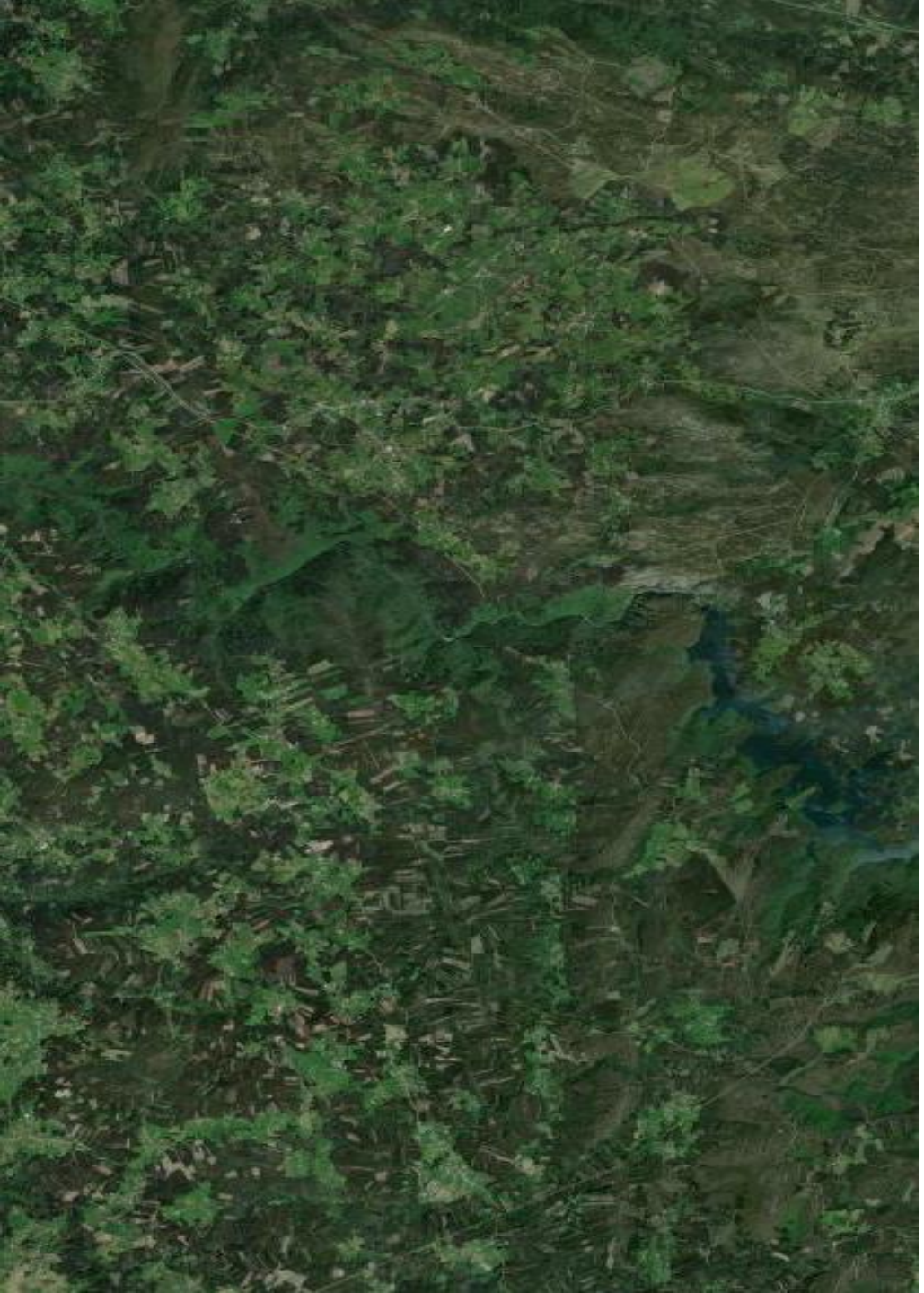


Assessing the Fragas do Eume Protected Area (NW Spain) Ecological Resilience to Climate Change through Projected Normalised Difference Vegetation Index

Lago-González, Pedro
Acuña-Alonso, Carolina
Álvarez, Xana
García-Ontiyuelo, Mario

19th sdewes Conference ROME 2024

2024, 8th to 12th September

Universidade Vigo

Ministerio de Ciencia, Innovación y Universidades
Cofinanciado por la Unión Europea
Agencia Estatal de Investigación

The corresponding autor

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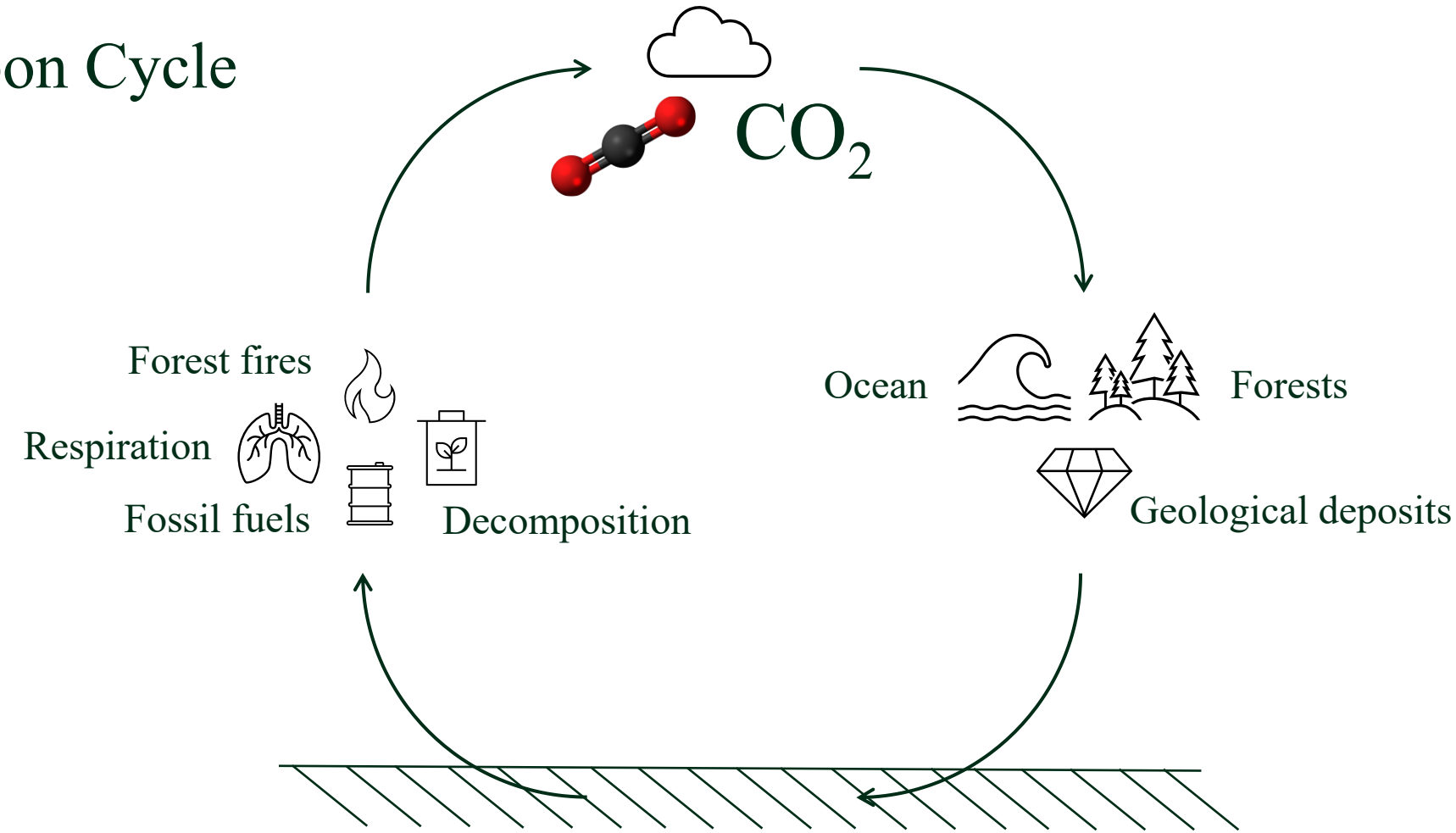


Predoctoral Researcher
Development of Machine Learning methods to assess the impacts of Climate Change and anthropogenic pressure in forest ecosystems

Hydroforestry Geomodelling Group
Forest Engineering School, Universidade de Vigo

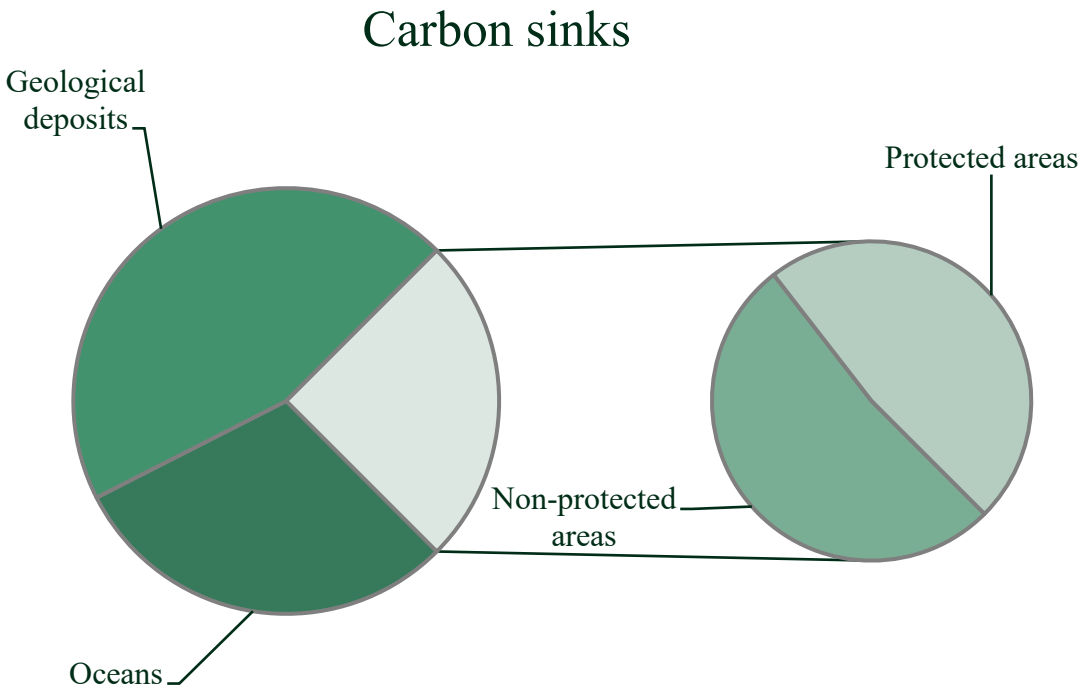
B.Sc. on Environmental Science
M.Sc. on Geophysics and Meteorology

The Carbon Cycle





Protected areas



Threats to biodiversity and the ecosystem health

Anthropogenic pressure

Urbanisation

Agriculture

Resource explotation

Tourism

Pollution

Rural abandonment

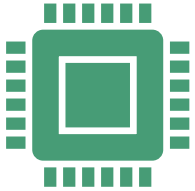
Climate change

Hotter weather

Temperature and precipitation extremes

Changes in the seasons

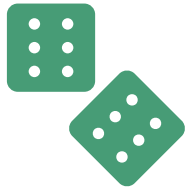
Objectives



Assess the vegetation health in Protected Areas in the short and long term future

NDVI

Machine learning



Prediction under future scenarios

SSP2-4.5

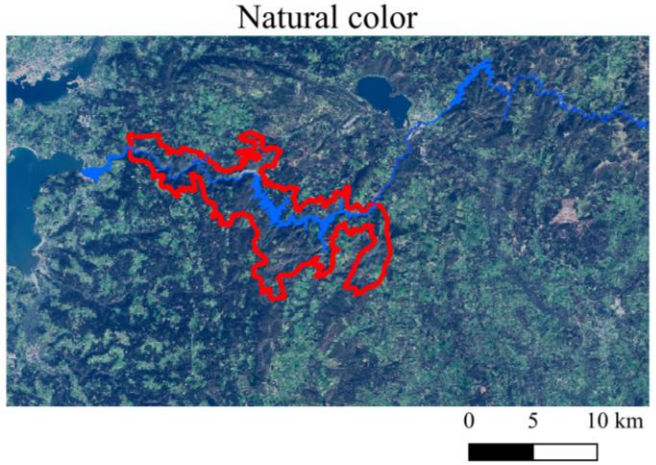
SSP5-8.5



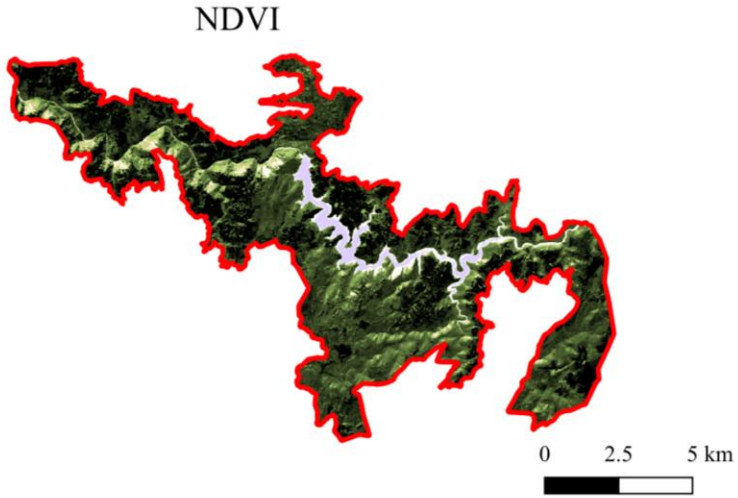
Aiding the decision-making process for a sustainable development

Area of Study

- Natura 2000 Network
  - SCI
  - Natural Park
- Example of climax vegetation in a temperate rainforest







The weather variables

WorldClim.org

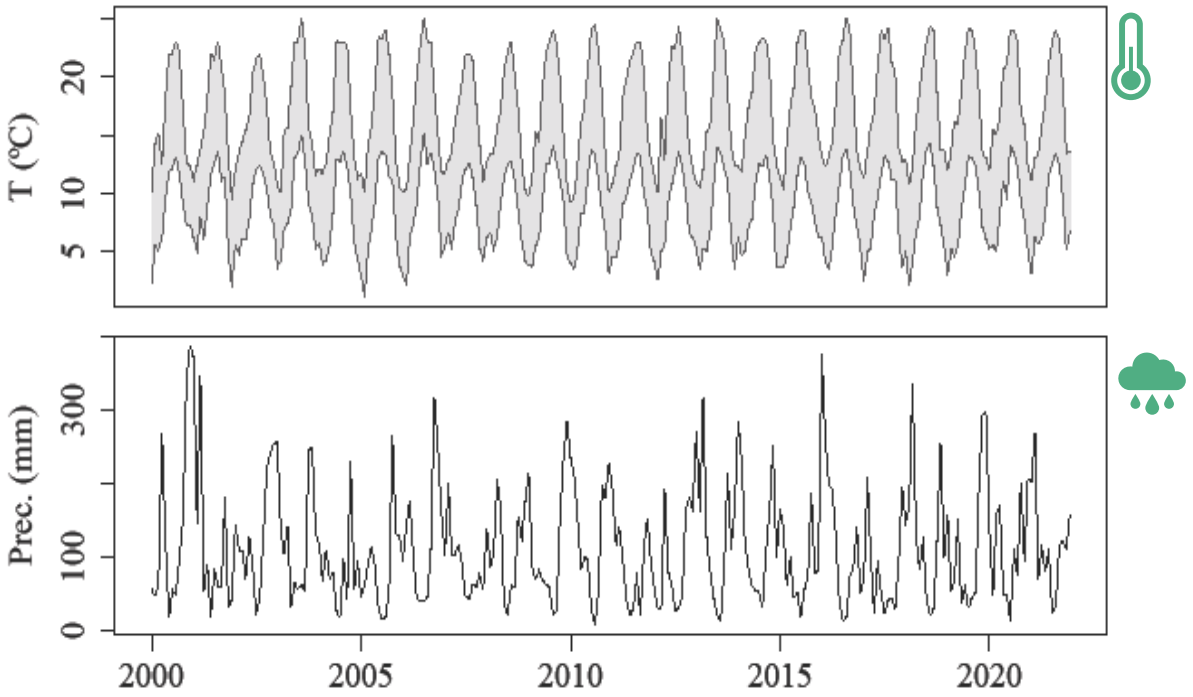
From 2000 to 2021

Resolution

Monthly
2.5’ × 2.5’

Variables

Temperature
Precipitation



The target variable

NASA EOSDIS LP DAAC: MODIS

From 2000 to 2021

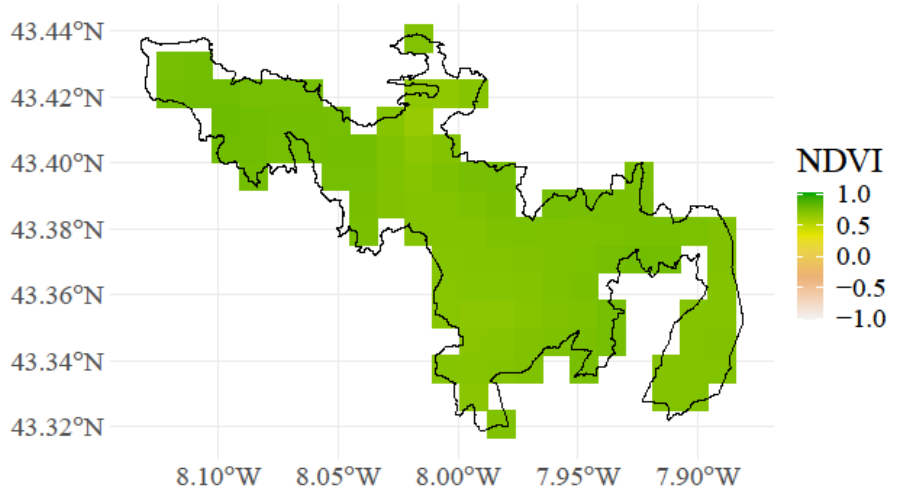
Resolution

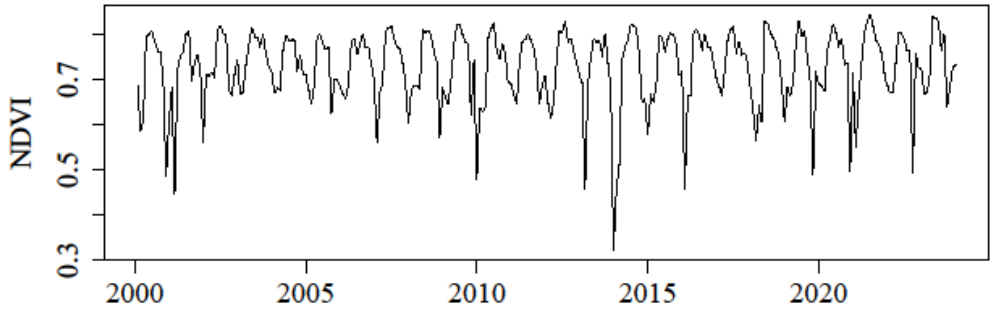
16 day

1 km × 1 km

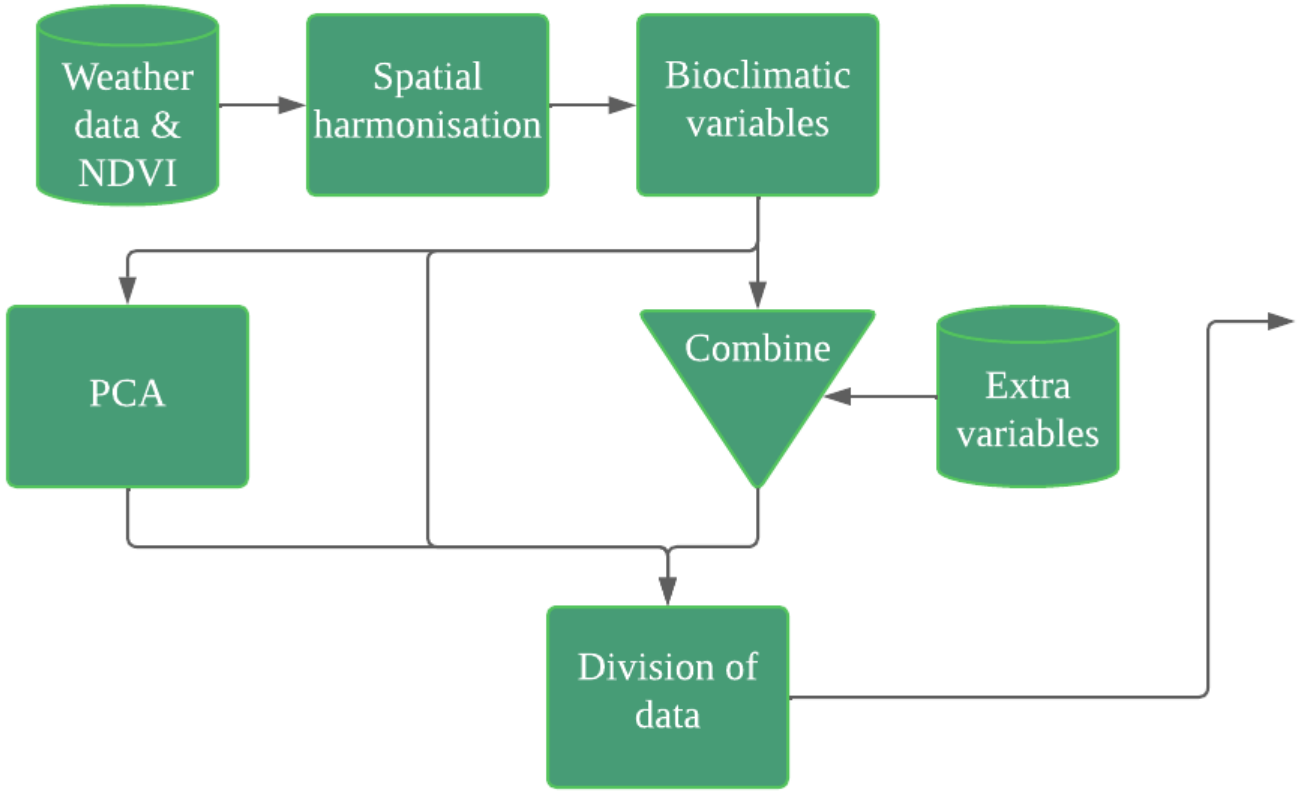
Variables

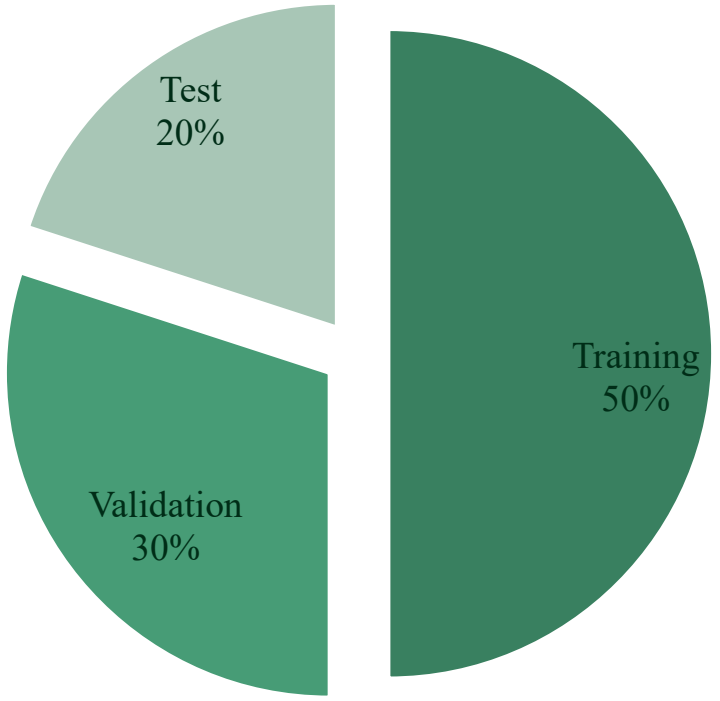
$$NDVI = \frac{NIR - R}{NIR + R}$$





Data pretreatment

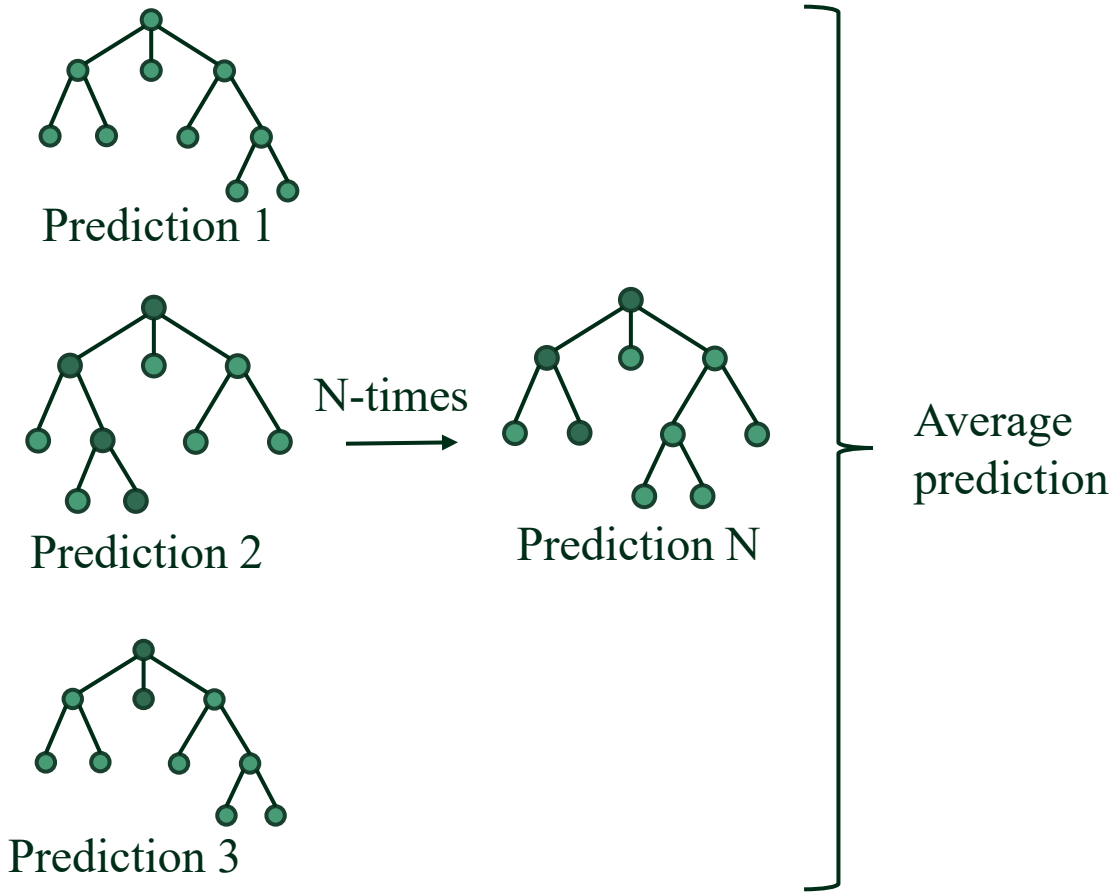




Random Forest

Hyperparameter tuning

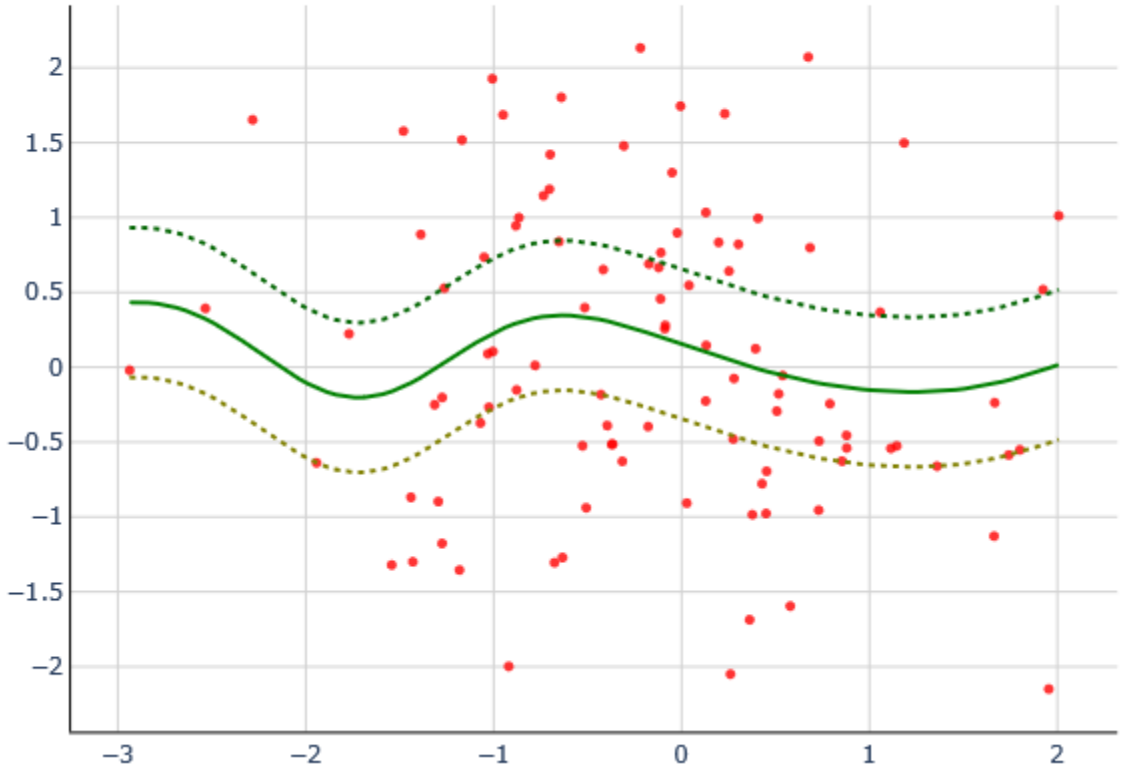
Hyperparameter	Range	Steps
# trees	[1,100]	99
Max depth	[-1,100]	101
Apply prepruning	Y/N	-



SVM

Hyperparameter tuning

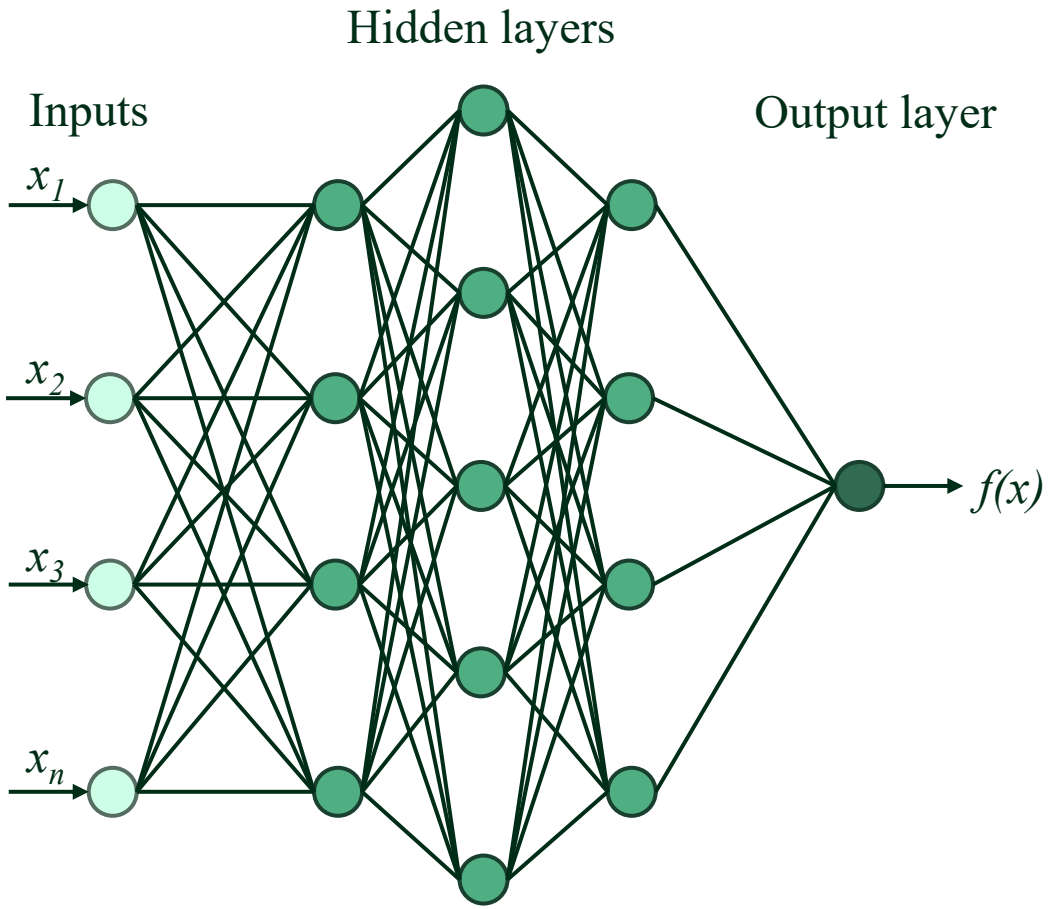
Hyperparameter	Range	Steps
SVM type	• υ-SVR • ε-SVR	-
γ	[3 10^{-5}, 8]	18, logarithmic
C	[0.03, 3.3 10^{4}]	20, logarithmic
Normalisation	Y/N	-
Kernel	RBF	-



Deep Learning

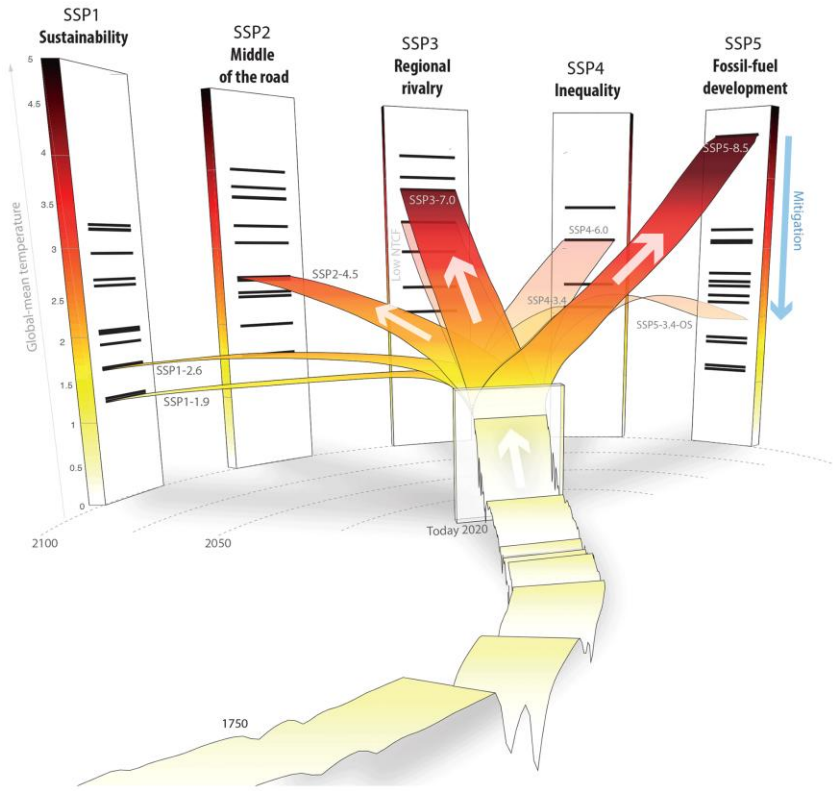
Hyperparameter tuning

Hyperparameter	Range	Steps
Activation function	• Tangent hyperbolic • ReLU • Max coordinate • Exponential ReLU	-
Epochs	[1, 100]	99
Learning rate	[0, 1]	10



Projections

Best fitting model



Souce: Cross-Chapter Box 1.4, Figure 1 in IPCC, 2021: Chapter 1. In: *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*

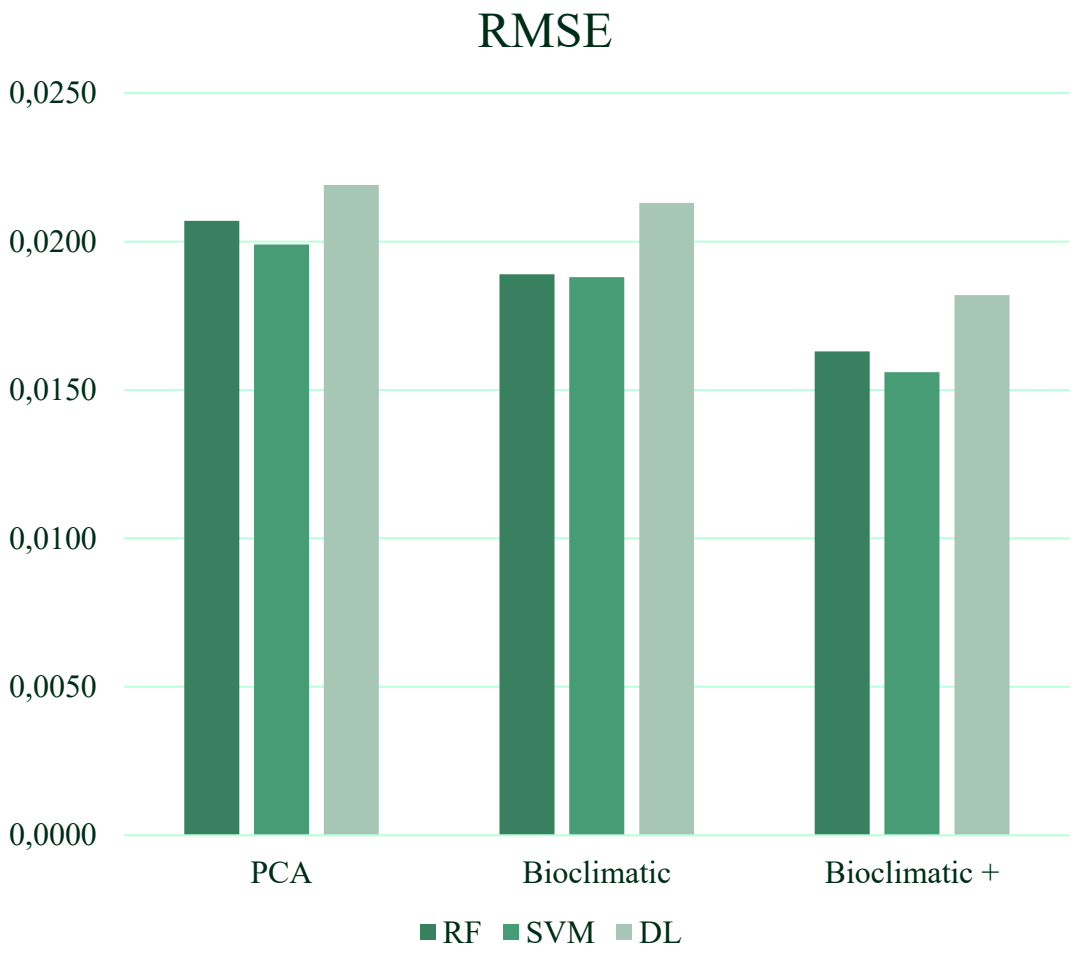
WorldClim.org

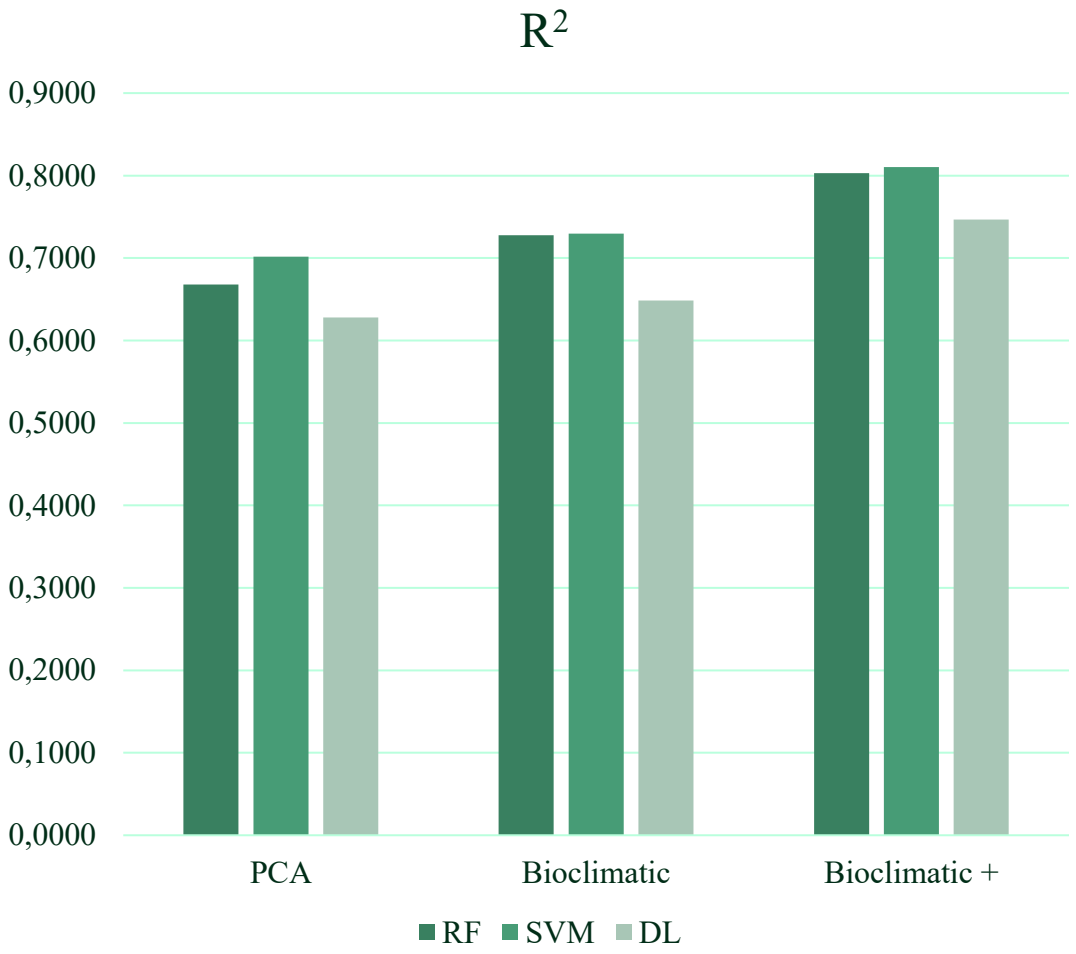
Periods
2021-2040
2081-2100

Resolution
20-year
30” × 30”

Variables
Temperature
Precipitation
Bioclimatic

Models performance





Projections

Best model: SVM

Hyperparameter	Optimal
SVM type	υ-SVR
γ	4.000
C	0.250
Normalisation	Yes
Kernel	RBF
Inputs	Bioclimatic +

Period	SSP2-4.5	SSP5-8.5
Present (2000-2021)	0.7268 ± 0.1152	0.7268 ± 0.1152
Short-term future	$0.7237 \pm 5.08\ 10^{-6}$	$0.7237 \pm 3.25\ 10^{-6}$
Long-term future	$0.7237 \pm 5.89\ 10^{-6}$	$0.7237 \pm 1.12\ 10^{-5}$

Conclusions

Models of accurate performance

RMSE ~ 0,015

R^2 ~ 0,8

Most optimalmdels

υ-SVR

Random Forest

Inputs: Bioclimatic +

Unvariability in the ecosystem health

Importance of managing PAs as a resilient agent against CC

Future works: Refine the model

Use a wider set of inputs (e.g. LULC, radiation, humidity)

Projections in the same temporal and spatial resolution

Aknowledgements

This research was funded by Grant TED2021-130241A-I00 funded by MCIN/AEI/10.13039/501100011033 and by the European Union NextGenerationEU/PRTR. This research was partially funded by the Conselleira de Educación, Universidade e Formación Profesional, Xunta de Galicia, Spain, under project GPC-ED431B 2022/12.

Universidade Vigo

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